

Marcos Talamantes

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EDUCATION

University of California, Merced

May 2026

Bachelor of Science in Mechanical Engineering, GPA: 3.1

Merced, CA

Relevant Coursework: Vibration & Controls, Mechatronics, Heat Transfer, Manufacturing, Aerospace Propulsion

Involvement: Kappa Sigma (Vice President), Institute of Electrical and Electronics Engineers (CAD Lead), SHPE (Member)

EXPERIENCE

E. & J. Gallo Winery - Supply Chain Engineering Intern

July 2026 - Present

- **Automation:** Leading an X-Ray inspection machine project to strengthen quality standards, improving detection to 2mm findings and increasing process output 2.2x over the original procedure.
- **Systems:** Utilized SAP Extended Warehouse Management (EWM) to track and manage product movement across warehouses, including damaged goods handling and inventory count support.
- **Process Improvement:** Collaborated with department leaders in a cross-functional Kaizen event to improve current Lean process, with the solution now in implementation.
- **Layout Design:** Drafted layouts in AutoCAD to reconfigure storage zones across 150,000 sq ft facility.

Woodpecker Microsystems - Engineering Intern

September 2025 - July 2026

- **Team Leadership:** Led a multidisciplinary team of students to develop a low-cost groundwater level monitoring system, designed for the Merced Irrigation system and validated in testing to 300 ft.
- **Solution:** Deployed a LiDAR based system achieving 0.42 ft accuracy well within 1 ft target, and integrated IoT communication with currently used transmission systems for real time groundwater analysis.
- **Evaluation:** Successfully created prototype with performance comparable to \$3K commercial sensors while reducing bill of materials (BOM) costs by 85%.
- **Result:** Selected to continue on the project post-capstone to refine the system for field deployment.

PERSONAL PROJECTS

Wildfire Protocol Sensor System | Arduino Microcontroller, BLE Communication, IoT Network, Sensors

July 2025

- **Sensor Integration:** Developed a dual-Arduino microcontroller system using DHT22, MQ-2 and water sensors for capturing continuous environmental condition monitoring measurements.
- **Data Logging & Visualization:** Implemented a BLE-based Circuit with RTC timestamping and SD logging to track on-site environmental trends and support early anomaly detection for proactive wildfire risk assessment.
- **Wireless Communication:** Integrated BLE module (HM-10) to transmit data to iOS and PC devices within a 40-meter range, enabling remote monitoring and alerts.
- **Enclosure Design:** Prototyped four variations of 3D printed casings, tested PLA enclosures under low-stress conditions and incorporated a backup coin cell to ensure uninterrupted data recording.

Deployable Aerial Reconnaissance Vehicle (DAR-V) UAV | Solidworks, Avionics Integration, Simulation

May 2025

- **Aerodynamic Design:** Utilized SOLIDWORKS simulation analysis on alternative airfoils, selecting Eppler 193 profile to maximize stability and lift-to-drag ratios for improvement in flight performance, fuel usage.
- **Avionics Integration:** Configured flight controller, electronic speed controller (ESC), and first person view system (FPV) with real time viewing camera and transmitter, enabling first-person UAV operation
- **Flight Control Software:** Programmed INAV for mixer/servo layouts and integrated GPS and compass sensors for coordinated flight control.
- **Additive Manufacturing:** Applied slicer software, print patterns, and filament selection to optimize 3D-printed UAV parts for flight performance in central valley conditions and aerodynamic loading.

SKILLS & CERTIFICATIONS

Certifications: Solidworks CAD Design Associate (CSWA), ID: C-VBWZE3L7Q4 (January 2026)

Technical: 3D Printing, Finite Element Analysis (FEA), Mechatronics, GNSS/GPS Survey

Languages & Software: SolidWorks, AutoCAD, MATLAB, SIMULINK, Python, Arduino IDE, SAP EWM, INAV (Drone Configuration), VS Code, SCORBASE, Excel, Microsoft Office